

A Three-Chamber Dynamic Release Device for Simultaneous Drug-Release Kinetics and *In Situ* Infection Challenge of Drug-Eluting Ultra-High-Molecular-Weight Polyethylene (UHMWPE) Joint Implants

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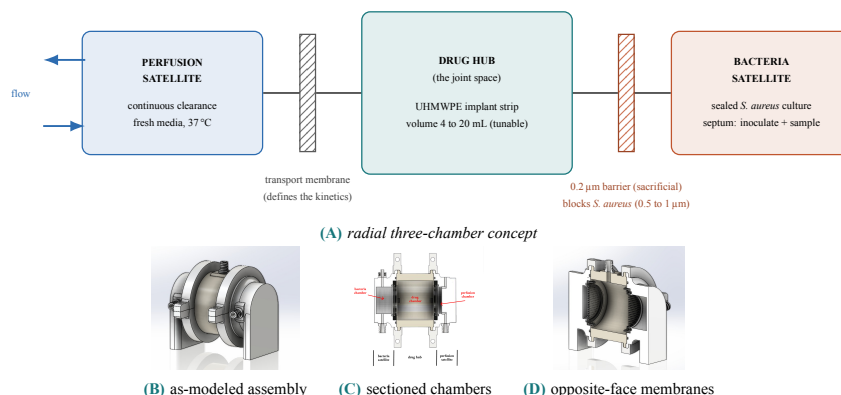


Figure 1. The DRD-3 three-chamber device. (A) Radial schematic: a central drug hub (joint space, holding the UHMWPE strip) is flanked on opposite faces by a perfusion satellite and a sealed bacteria satellite; placing the two membranes on opposite faces rather than in series keeps biofilm on the sacrificial 0.2 µm barrier from reaching the transport membrane that sets the kinetics. (B) As-modeled assembly with the C-ring wrappers that load each joint. (C) Top-down section through the three chambers. (D) Isometric cut showing the two opposite-face membranes in modular cartridges. Sealing uses three identical O-ring pairs (2.0 mm cord, 25% compression).

Introduction. Periprosthetic joint infection (PJI) remains one of the most serious complications after total joint arthroplasty. An otherwise successful reconstruction can be converted into a prolonged course of revision surgery and antibiotic treatment, leading to impaired function and increased patient morbidity.^{1,2} Although systemic antibiotics are routinely used in PJI treatment, achieving sustained therapeutic concentrations inside the joint is difficult because intraarticular drug availability is limited by local transport barriers, synovial fluid interactions, and the need to avoid toxic serum concentrations.^{1,3} Local antibiotic delivery from drug-eluting implants is therefore attractive because it can concentrate drug at the site of infection while reducing systemic exposure.

Ultra-high-molecular-weight polyethylene (UHMWPE) is the material of choice for load-bearing total joint implants, and antibiotic-eluting UHMWPE is being developed as a functional spacer or implant material for PJI treatment.^{1,2} Translation of these materials requires in-vitro models that capture the release, transport, and clearance conditions governing drug concentration inside the joint. The Dynamic Release Device (DRD) was recently developed to address this need by combining a drug chamber representing the joint space, a diffusion membrane representing synovial transport, and a perfused chamber representing clearance.³ However, the DRD evaluates drug transport without a live infection challenge, as the current design would mean adding bacteria directly to the drug chamber; this would allow biofilm to foul the transport membrane, thus altering the permeability that defines the measured release kinetics. In this study, we designed a three-chamber DRD-3 to preserve the DRD transport measurement while adding a physically isolated *Staphylococcus aureus* challenge in the same experiment, furthering our understanding of intrarticular dynamics by simulating the infection-based scenario where a drug-eluting antibiotic would be most useful.

Methods. We designed a clean-sheet, three-chamber device (the DRD-3) in CAD (Fig. 1A; assembly in Fig. 1B). A central drug hub represents the joint space, holds the UHMWPE implant strip, and targets a tunable fluid volume of 4 to 20 mL spanning healthy to infected knee synovial volumes. The hub is flanked on opposite faces by two satellites: a perfusion satellite providing continuous fresh-media clearance (the validated transport physics, preserved) and a bacteria satellite housing a sealed culture (Fig. 1C). Because the two membranes sit on opposite faces of the hub rather than stacked in series (Fig. 1D), neither membrane is exposed to the other's fouling. Each membrane is sealed off-device inside a modular lid-and-base cartridge; a single tapered tooth (tapering from 2.0 to 1.0 mm) both captures the membrane sandwich and retains the finished cartridge in its satellite, and parallel protective bars shield the membrane from the UHMWPE strip without restricting diffusion. Every sealing interface uses one identical static O-ring specification (2.0 mm cord, 1.5 mm gland, 0.5 mm stand-proud giving 25% compression), so the entire two-membrane device seals on three identical O-ring pairs. Each satellite is drawn to the hub by a two-piece C-ring wrapper whose U-channel applies uniform radial load around the joint with one bolt per side, replacing point-loaded flange bolts. Bacterial isolation is enforced by a sacrificial 0.2 µm microfilter on the bacteria-facing membrane that passes drug freely but physically blocks *S. aureus* (0.5 to 1 µm) from migrating toward the transport membrane; the culture is accessed through a self-sealing septum for inoculation and colony-forming-unit (CFU) sampling without breaching sterility. Structural parts are stereolithography (SLA) resins (Accura-60-class chassis; Somos 9120 for the compliant inner cartridge parts) with medical-grade silicone/EPDM (USP Class VI) O-rings. A three-phase validation protocol is defined: Phase 0, a dye-perfusion leak test at 64 mL/h for 24 h (acceptance below 2% volume loss per 24 h); Phase 1, reproduction of the published release kinetics across flow rate (2 to 64 mL/h), temperature (room tem-

perature versus 37 °C), and membrane thickness; and Phase 2, introduction of *S. aureus* in the bacteria satellite with CFU sampling over 24 h to confirm zero bacterial migration to the perfusion side. As a non-clinical benchtop instrument, this work involves no human or animal subjects, and sex as a biological variable is not applicable at this stage.

Results. A complete three-chamber DRD-3 was realized in CAD. The design consolidates all sealing to three identical O-ring pairs and a single tapered-tooth feature that performs both membrane capture and cartridge retention, and it replaces the earlier multi-bolt, point-loaded clamp with one continuous C-ring wrapper per satellite (four wrappers, one bolt per side) that distributes radial load uniformly around each joint. The 0.2 µm isolation barrier is sized well below the *S. aureus* cell diameter, and the drug hub is parameterized so hub depth can be tuned to a target volume. The as-modeled internal volume is being extracted from CAD to confirm it lands in the 4 to 20 mL window [measured volume, mL]. Experimental validation is in progress: Phase 0 leak loss [value % / 24 h]; Phase 1 agreement of DRD-3 half-life trends with the published baseline [result]; Phase 2 bacterial count on the perfusion side [CFU; expected 0].

Discussion. The defining feature of the DRD-3 is the radial, opposite-face, two-membrane architecture, which decouples the infection challenge from the transport measurement. Biofilm is permitted, and can optionally be imaged through a swappable borosilicate window, on the sacrificial barrier, while the kinetics-defining membrane stays clean. The modular design targets the reproducibility and assembly failure modes of press-fit SLA stacks: the membrane is sealed as a bench-built module, the seal is an elastomer rather than a flexing printed feature, and the wrapper clamp removes press-fit interfaces from the load path. Limitations are that the as-modeled internal volume is not yet verified against the 4 to 20 mL target; the experimental validation (leak integrity, kinetics reproduction, and isolation) is ongoing; SLA parts must be fully post-cured and washed before live-culture runs to avoid residual-monomer antimicrobial bias; and the bacteria and perfusion chamber volumes are currently asymmetric, which is under review. The mapping to anatomy is also deliberately partial: the drug hub represents the intraarticular space and the perfusion satellite the joint's clearance pathway, but the isolated bacterial compartment is an engineering abstraction rather than an anatomical analog (in vivo the infection resides within the joint space itself), adopted so that biofilm cannot foul the transport membrane that defines the kinetics. Within the framework of our objective, the device is built so that, once it reproduces the published transport baseline, it can add a live antimicrobial-efficacy readout without compromising that measurement.

Significance. This platform would allow drug-eluting UHMWPE bearing materials to be screened for both intraarticular release kinetics and live antimicrobial efficacy on a single benchtop, directly supporting the development of local-delivery strategies against periprosthetic joint infection.

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